IBM SkillsBuild AICTE Internship

Data Analytics with AI — Submission Report

Student Work & Prompt Workbook Evidence

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Submission Date	05 October 2026

Submission note: This report documents the supplied internship resources and the completed Data Analytics prompt-work evidence. It does not claim completion of recordings or activities for which no completion evidence was supplied.

1. Internship / Resource Overview

The supplied internship resource document covers IBM SkillsBuild Data Analytics with AI, including learning resources, masterclass sessions, recordings, practice material, and project-related resources.

2. Submission Components

• Student details and submission information.

• Data Analytics prompt workbook evidence (Prompts 1–4).

• Prompt 1: dataset structure and column meanings.

• Prompt 2: data-quality checks using the raw dataset.

• Prompt 3: Customer_Segment classification formula.

• Prompt 4: management business questions for analysis.

3. Prompt 1 — Dataset Structure

The following table describes the original dataset fields shown in the workbook.

Column	Meaning	Data Type	Business Use
Order_ID	Unique order identifier	Identifier	Track individual orders
Order_Date	Date when the order was placed	Date	Analyze sales trends
Customer_ID	Unique customer identifier	Identifier	Analyze customer behavior
Product	Product purchased	Categorical	Compare product performance
Category	Product category	Categorical	Category-wise analysis
Region	Region of order	Categorical	Regional performance
Quantity	Units ordered	Numerical	Analyze demand
Revenue	Sales generated	Numerical	Measure sales
Profit	Profit generated	Numerical	Measure profitability

4. Prompt 2 — Data Quality Check

The workbook identifies these data-quality checks. Raw_Data is retained unchanged; issues are flagged for review rather than silently corrected.

Check	Result	Status
Missing Customer_ID	26	Needs review
Missing Quantity	23	Needs review
Missing Region	16	Needs review
Negative Quantity	41	Needs review
Negative Profit	15	Investigate
Duplicate Order_ID rows	0	Verify
Revenue stored as text	0	Standardize later

Additional observations in the workbook include inconsistent capitalization for Category/Region, mixed Revenue formats, and suspicious date values that should be verified against the source before correction.

5. Prompt 3 — Customer Segment

Customer_ID is in column C. A customer is classified as New Customer when the ID appears once and Repeat Customer when it appears more than once.

Formula: =IF(COUNTIF(C2:$C,C2)=1,"New Customer","Repeat Customer")

The formula can be entered in the Customer_Segment column and filled down for all rows.

6. Prompt 4 — Management Business Questions

The workbook defines 12 measurable questions using the available sales, product, customer and regional fields.

Category	Business Question	Columns Needed
Sales	Which month generated the highest total Revenue, and how did monthly Revenue change across the year?	Order_Date, Month, Revenue
Sales	Which month generated the highest total Profit, and which generated the lowest?	Order_Date, Month, Profit
Sales	Which Year/Month combination has the highest order-value mix?	Order_Date, Month, Year, Order_Value_Category, Order_ID
Product	Which Product generated the highest total Revenue?	Product, Revenue
Product	Which Product generated the highest total Profit and which generated the lowest?	Product, Profit
Product	Which Category has the highest total Quantity sold?	Category, Quantity
Customer	Which customers placed the most orders, and are they classified as Repeat Customers?	Customer_ID, Order_ID, Customer_Segment
Customer	Which Customer_Segment generates more total Revenue: New Customer or Repeat Customer?	Customer_Segment, Revenue
Customer	How many high-value orders fall into each Customer_Segment?	Customer_Segment, Order_Value_Category, Order_ID
Region	Which Region generated the highest total Revenue and which generated the lowest?	Region, Revenue
Region	Which Region generated the highest total Profit and which had the lowest/negative profitability?	Region, Profit
Region	Which Region has the highest average Quantity per order, and how does that compare with its Revenue and Profit?	Region, Quantity, Revenue, Profit, Order_ID

7. Final Submission Checklist

• Student name, roll number, college, program, email and submission date verified.

• Prompt 1–4 answers included in the accompanying Excel workbook.

• Raw_Data retained unchanged for the data-quality review.

• Excel workbook checked for readable formatting and wrapped text.

• PDF report checked for page layout and table overflow.

• Supermarket Sales Analysis project evidence: attach separately only if actually completed and evidence is available.

• Screenshots/evidence of completed learning activities: attach only where required and available.

8. Student Details — Fill Before Final Upload

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Final note: The accompanying Excel workbook is the detailed evidence for the four Data Analytics prompts. Upload any other required deliverables only when completion evidence is available.